Data Analytics Internship – Task 1

Sales Performance Analysis

Submitted to: Alfido Tech

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Domain: Data Analytics Internship
Tools Used: Python, Pandas, Matplotlib, Google Colab

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1. Introduction

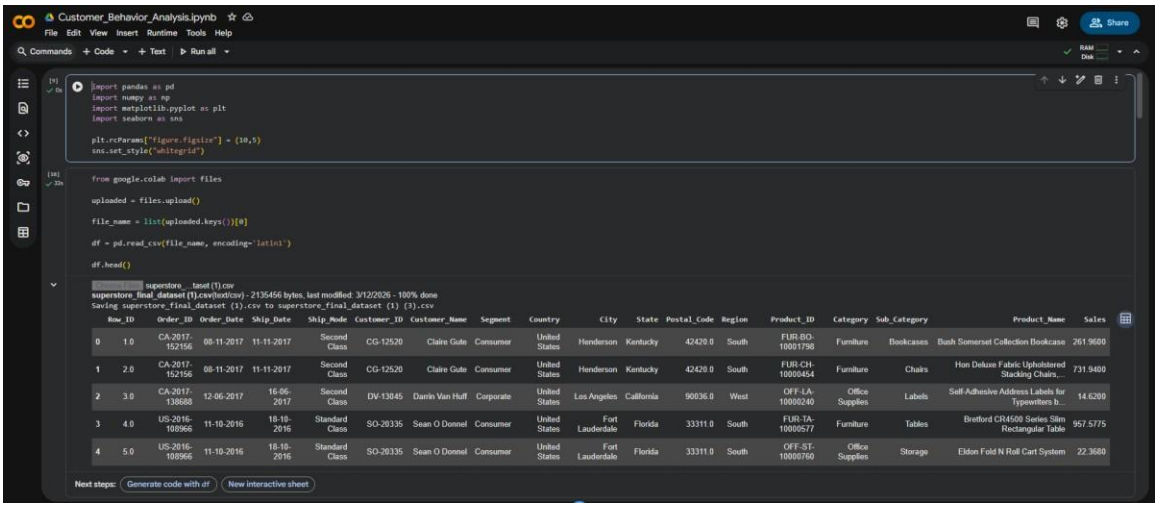
Data analytics is an important process that helps organizations understand patterns, trends, and insights from large amounts of data. By analyzing data effectively, businesses can make better decisions, improve their strategies, and enhance customer satisfaction. In today's digital world, companies rely on data-driven insights to understand customer behavior, identify market trends, and improve overall business performance.

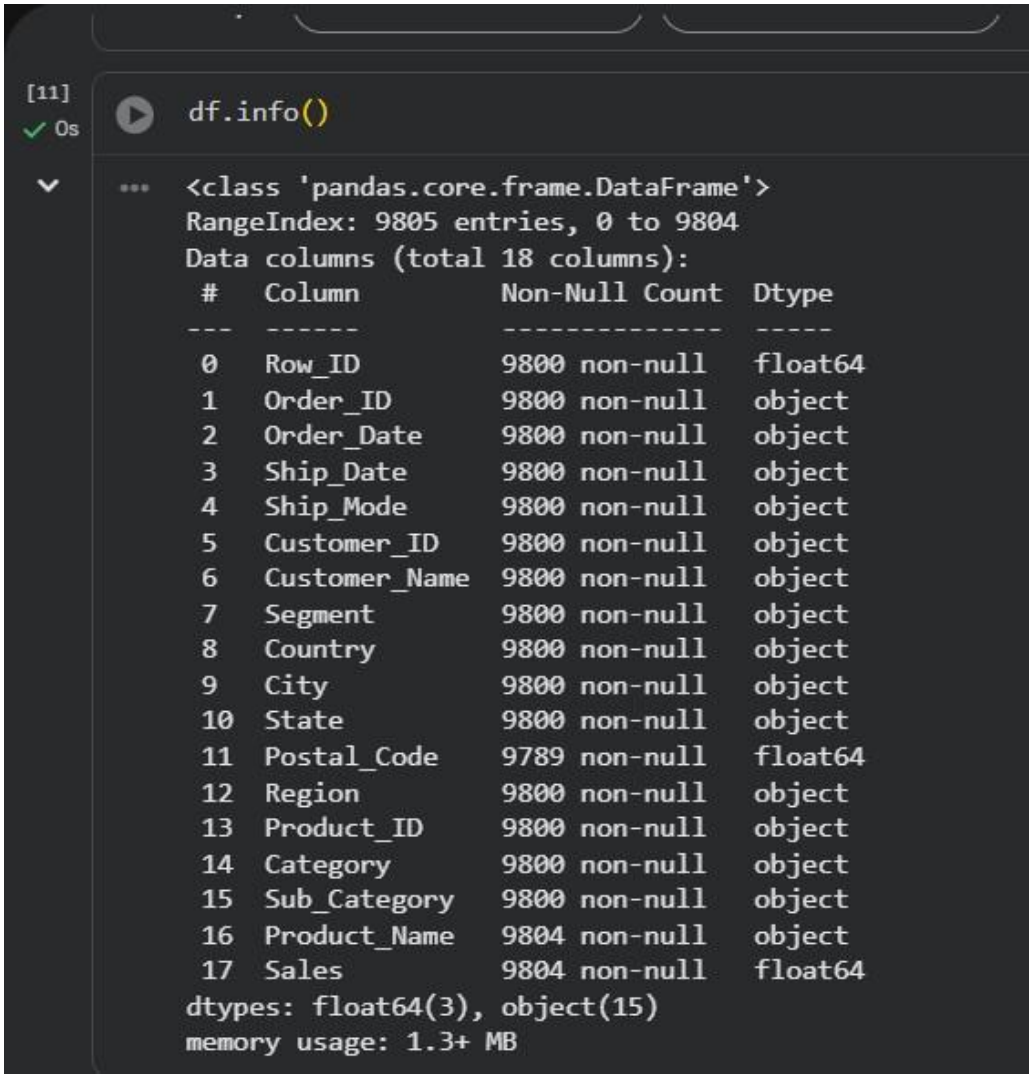
The objective of this task is to analyze a sales dataset to understand customer purchasing behavior and sales performance across different categories and regions. The dataset contains information about orders, customers, product categories, and sales values. By analyzing these variables, it becomes possible to identify important patterns that can help businesses improve their marketing strategies and product offerings.

The analysis was performed using Python in Google Colab. Various exploratory data analysis techniques and visualizations were used to identify trends and patterns in the dataset.

2. Dataset Overview

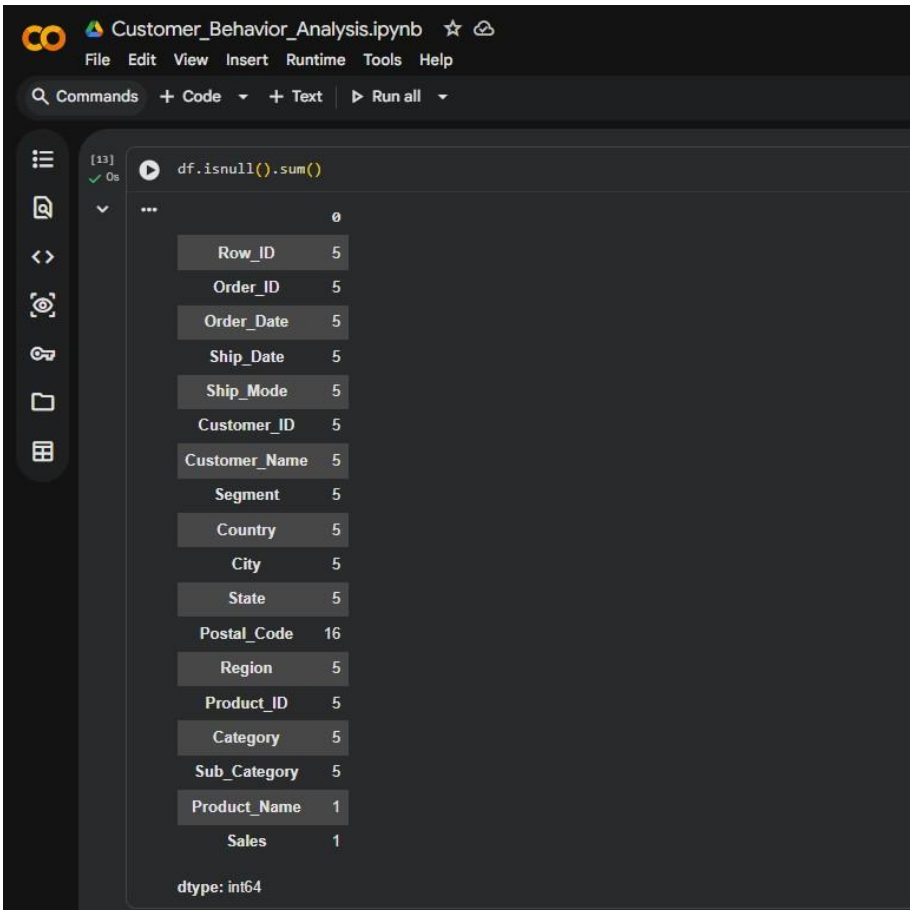
The dataset contains sales transaction data including information about orders, customers, products, and regions. Important columns include Order ID, Customer Name, Category, Region, and Sales.

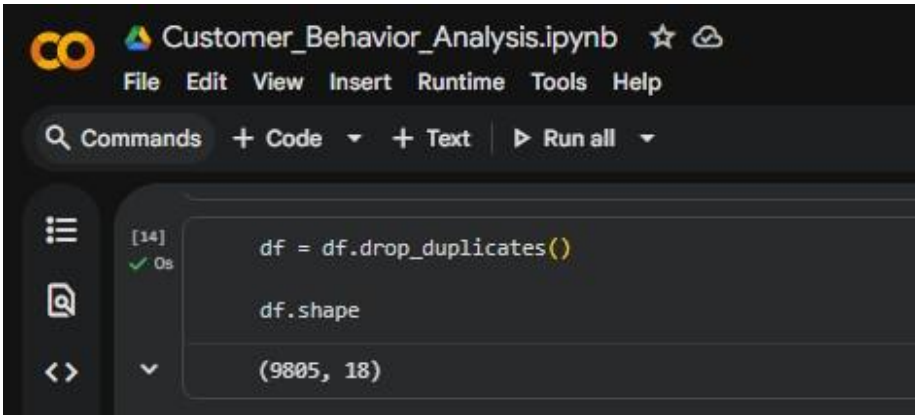




3. Data Cleaning

Data cleaning was performed to ensure data quality. Missing values were checked and duplicates were removed.





4. Summary Statistics

Summary statistics were calculated to understand the distribution of the dataset.

[12]
0s

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df.describe()
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	Row_ID	Postal_Code	Sales
count	9800.000000	9789.000000	9804.000000
mean	4900.500000	55273.322403	230.705325
std	2829.160653	32041.223413	626.534737
min	1.000000	1040.000000	0.000000
25%	2450.750000	23223.000000	17.246000
50%	4900.500000	58103.000000	54.485000
75%	7350.250000	90008.000000	210.605000
max	9800.000000	99301.000000	22638.480000

5. Data Visualization

[15]
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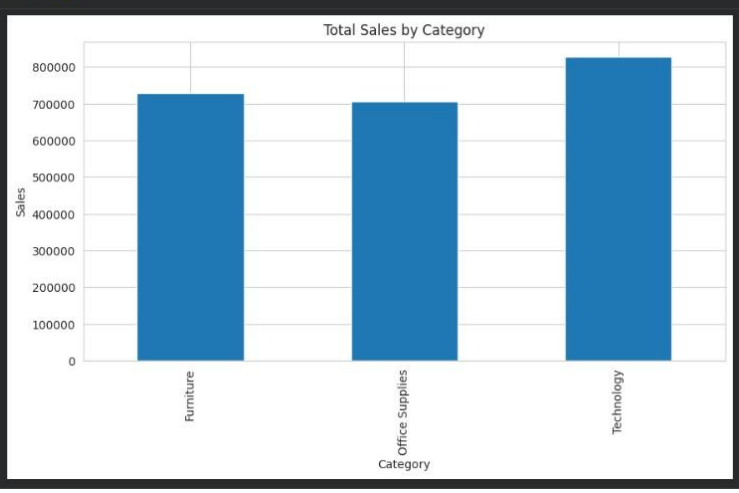
```
category_sales = df.groupby("Category")["Sales"].sum()

category_sales.plot(kind="bar")

plt.title("Total Sales by Category")
plt.ylabel("Sales")
plt.show()
```



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[16]
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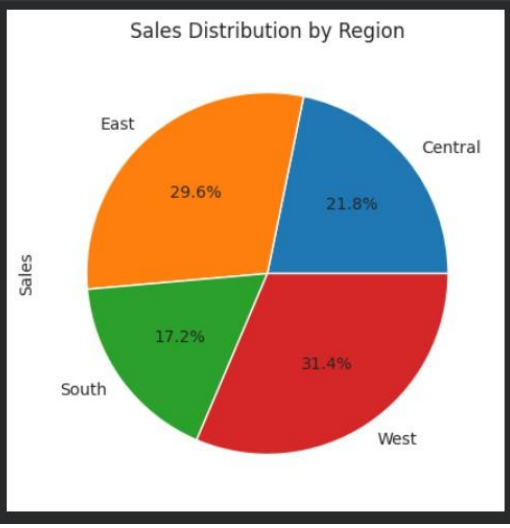
region_sales = df.groupby("Region")["Sales"].sum()

region_sales.plot(kind="pie", autopct="%1.1f%%")

plt.title("Sales Distribution by Region")
plt.show()
```

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[20]
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top_products = df.groupby("Product_Name")["Sales"].sum().sort_values(ascending=False).head(10)

top_products.plot(kind="bar")

plt.title("Top 10 Products by Sales")
plt.ylabel("Sales")
plt.show()
```

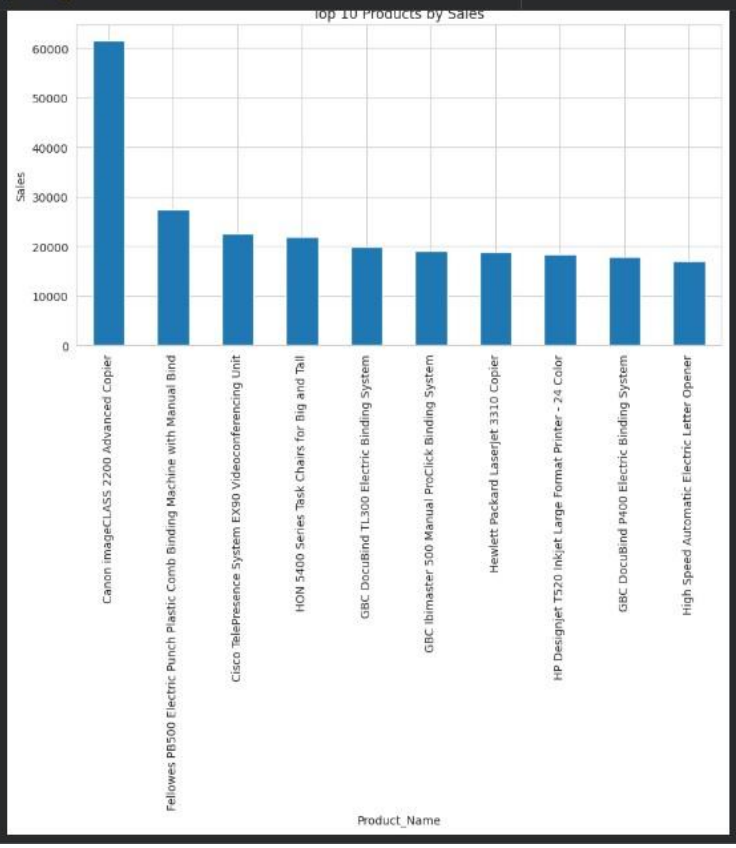

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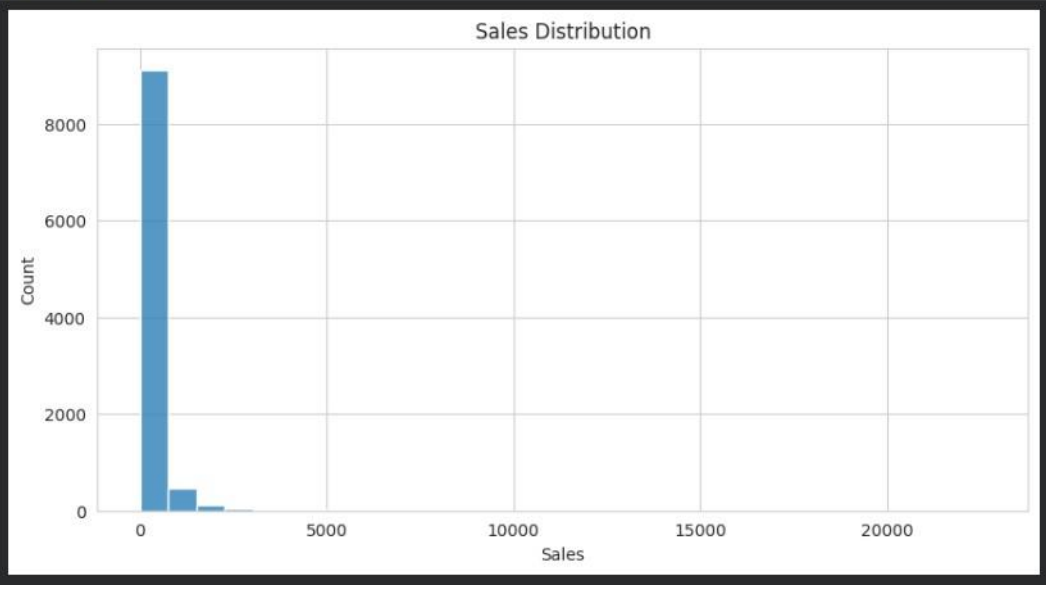
sns.histplot(df["Sales"], bins=30)

plt.title("Sales Distribution")
plt.show()
```

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[21]
segment_sales = df.groupby("Segment")["Sales"].sum()

segment_sales.plot(kind="bar")

plt.title("Sales by Customer Segment")
plt.show()
```

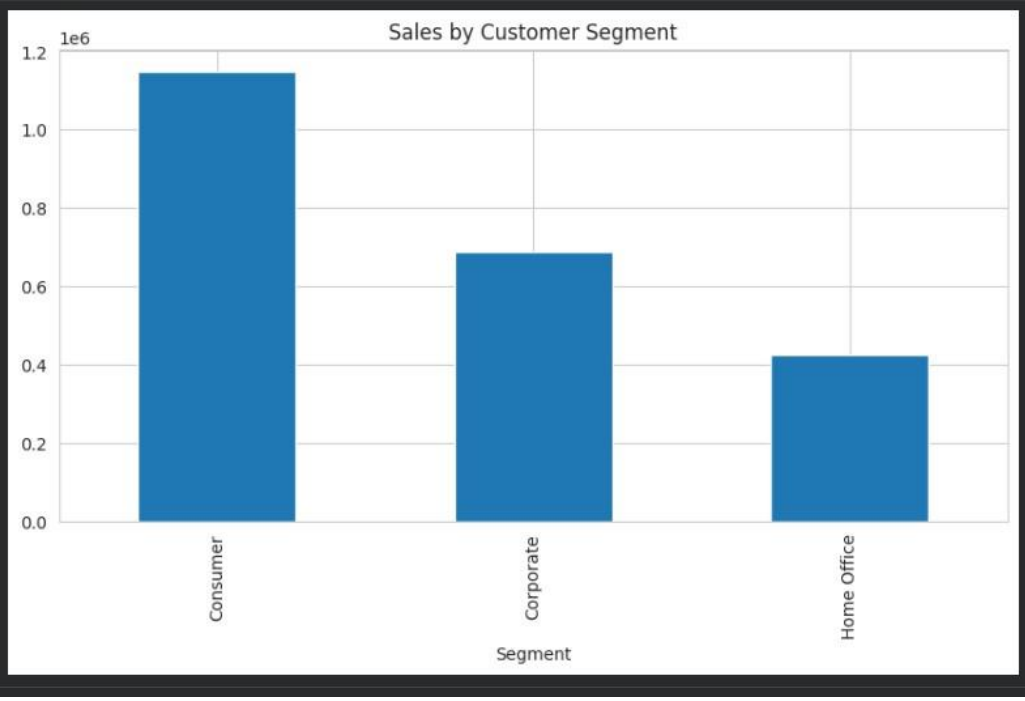

6. Key Insights

1. Technology category generated highest sales.
2. Few products contribute most revenue.

3. Sales vary across regions.
4. Most sales are low-value.

7. Recommendations

1. Focus on high-performing categories.
2. Promote top products.
3. Improve low-performing regions.
4. Introduce loyalty programs.
5. Bundle products.

8. Executive Summary

This project analyzed sales data to identify patterns and trends. Visualizations helped understand category performance and regional distribution. Recommendations were provided to improve business strategies.

9. Project Links

https://github.com/Swayamchavan/InternSpark-Internship